2019秋高数上期中试题(回忆版).

1. Determine whether the following statements are true or false? No justification is necessary.

(1) If $f'(x)$ is bound on $(0,1)$, so is $f(x)$.

(2) Let $f(x)$ be defined on $(-\infty,+\infty)$. There must be a local maximum point of $f(x)$ between two local minimum points of $f(x)$.

(3) If $f(x)$ is differentiable on $(-1,1)$, and $f(-1)=f(1)$, then $f'(c)=0$ for some number $|c|<1$.

(4) If $f(x)$ is a continuous, even function on $[-1,1]$, then $g(x)=\int_0^x f(t)\,dt$ is odd and differentible on $[-1,1]$.

(5) If $f(x)$ is a continuous, periodic function on R (T is the period), then $g(x)=\int_0^x f(t)\,dt$ is also a periodic function with the period T.

2. Multiple Choice Questions: (only one correct answer for each of the following questions.)

(1) Which of the following functions is not differentiable at $x=0$?

(A) $|x|\sin|x|$ (B) $|x|\sin(\sqrt{|x|})$ (C) $\cos|x|$ (D) $\cos\sqrt{|x|}$

(2) Suppose that $f(x)$ is differentiable at $x=0$ and $f(0)=0$. Then $\lim\limits_{x\to 0}\frac{x^2f(x)-2f(x^3)}{x^3}=$

(A) $-2f'(0)$ (B) $-f'(0)$ (C) $f'(0)$ (D) 0

(3) Suppose that $f(x)$ has a second derivative and $f'(0)=0$, $\lim\limits_{x\to 0}\frac{f''(x)}{x}=1$. Then

(A) $f(0)$ is a local minimum value.

(B) $f(0)$ is a local maximum value.

(C) $(0,f(0))$ is a point of inflection of the curve.

(D) $(0,f(0))$ is neither a local extrema nor a point of inflection of the curve.

(4) Suppose that $f(x)$ is defined on $(-\infty,+\infty)$. Which of the following statements is equivalent to the statement that "$f(x)$ is differentiable at $x=a$"?

(A) $\lim\limits_{h\to 0}(f(a+h)+f(a-h)-2f(a))=0$. (B) $\lim\limits_{h\to 0}\frac{f(a+h)-f(a-h)}{2h}$ exists.

(C) $\lim\limits_{h\to 0}\frac{f(a+h^2)-f(a)}{h^2}$ exists. (D) $\lim\limits_{h\to 0}\frac{f(a+h^3)-f(a)}{h^3}$ exists.

(5) Suppose that $f(x)>0$, $f'(x)>0$, and $f''(x)>0$ for all $x\in[a,b]$. Let $M=\int_a^b f(x)\,dx$, $N=f(a)(b-a)$, and $P=\frac{f(a)+f(b)}{2}(b-a)$. Then

(A) $N<P<M$ (B) $N<M<P$ (C) $M<N<P$ (D) $M<P<N$.

3. Let $f(x)=\frac{x^3}{x^2+1}$.

(1) Identify the inflection points and local maxima and minima of the function that may exist.

(2) Identify the horizontal, vertical, or oblique asymptotes that may exist.

(3) Graph the function.

4. Find the limits.

(1) $\lim\limits_{x\to 1}\left(\frac{\sin 5x}{x}+\frac{x^3+x^2-2}{x^2+2x-3}\right)$

(2) $\lim_{n\to\infty}\frac{1}{n}\left(\sqrt{1-(\frac{1}{n})^2}+\sqrt{1-(\frac{2}{n})^2}+\cdots+\sqrt{1-(\frac{n}{n})^2}\right)$

5. Evaluate the definite integral.

(1) $\int_{-1}^{1}|a-t|t^2\,dt$, where $a\in(-1,1)$.

(2) $\int_0^{\pi}\frac{\sin 2x}{\sqrt{1-\cos x}}\,dx$.

6. Find the volume of the solid generated by revolving the region bounded by $x=12(y^2-y^3)$ $(0\le y\le 1)$ and y-axis about the line $y=2$.

7. Use the linear approroximation of $f(x)=\tan x$ at $a=\frac{\pi}{6}$ to estimate the value of $\tan\frac{11\pi}{60}$. Comparing the estimation with the true value, which one is larger?

8. Find the area of the region in the first quadrant bounded on the left by the y-axis, below by the curve $x=2\sqrt{y}$, above left by the curve $x=(y-1)^2$, and above right by the line $x=3-y$.

9. Let $F(x)=\int_{2019}^{x^2}\cos(2t^2)\,dt$. Find all the critical points for $F(x)$ on $[-1,1]$.

10. (Use Rolle's theorem to prove the mean value theorem.) If the function $f(x)$ is continuous on $[a,b]$, and differentiable on (a,b), prove that there exists a number c in (a,b), such that

$$f'(c)=\frac{f(b)-f(a)}{b-a}.$$